

GC-8 (Cal portion): Application Targets — Tractability Assessment

Cal (Claude 4.7)

May 12, 2026

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Purpose

GC-4 (Grace) surveyed solved problems and found ~33% are GC-amenable. GC-8 (this note) projects forward: identify *open* problems amenable to the GC method, with tractability ratings to guide which deserve dedicated research programs.

Tractability scale (1-5): - 1: Highly tractable. Short program (weeks to months). - 2: Tractable. Moderate effort (6-12 months). - 3: Challenging. Multi-year program with clear path. - 4: Hard. Multi-year program with uncertain path. - 5: Research frontier. May not be GC-amenable in current formulation.

Grace's portion catalogs the targets and their formal statements. This portion assesses *whether GC applies* and *how hard it would be*.

The Three GC Necessary Conditions (from GC-4)

For a problem to be GC-amenable, GC-4 identified three requirements:

1. Finite classification of candidates (or finite parameterization).
2. Independent bounds meet with zero room (the constraint pair).
3. Uniqueness rather than existence (“which one” not “does one exist”).

Each target below is rated against these three.

Tier 1: Highly Tractable (Rating 1-2)

Target A: Sphere Packing in Other Dimensions

Statement: For which dimensions n is the optimal sphere packing density known and uniquely realized?

GC-amenability: HIGH (rating 1). - Finite candidate space: lattice packings in given dim are countable, often classifiable. - Independent bounds: linear programming upper bound (Cohn-Elkies) vs known lattice constructions (lower bound). - Uniqueness: when bounds meet, packing is unique up to symmetry.

Status: Viazovska's 2016/2019 work proved dim 8 (E₈) and dim 24 (Leech) by finding magic LP functions. Same method may extend to other dimensions.

Key constraint pair: LP bound (upper) meets lattice density (lower).

Tractable extensions: - Dim 16 (the Barnes-Wall lattice — open whether optimal) - Dim 32 (BW₃₂ — open) - Lower dimensions where the bound is known but uniqueness is open

Risks: - The Viazovska magic functions exist only for specific dimensions. Constructing them in other dims may fail. - Some dimensions may have non-lattice optimal packings (e.g., dim 10), complicating uniqueness.

Recommended action: Direct GC application. Methodology paper can cite this as a clean GC target.

Target B: Error-Correcting Code Boundaries

Statement: Identify codes that meet known bounds (Hamming, Singleton, Plotkin, Linear Programming, etc.) with equality.

GC-amenability: VERY HIGH (rating 1). - Finite candidate space: codes of given length and rate over fixed alphabet are finite. - Independent bounds: Hamming sphere bound vs construction lower bound. LP bound vs algebraic construction. - Uniqueness: when bounds meet, codes are “perfect” — unique up to permutation.

Status: The classification of perfect codes is essentially complete (Hamming, Golay codes). MDS codes meeting Singleton bound are well-studied (Reed-Solomon). Many open questions remain about which intermediate-distance bounds are tight.

Key constraint pair: information-theoretic upper bound vs constructive lower bound.

Tractable extensions: - Quantum error correction: tradeoffs between code distance, rate, and quantum codes (CSS bound, quantum Singleton bound). - Codes on non-Hamming metrics: rank-metric codes (Gabidulin), Lee-metric codes. - Locally recoverable codes (LRC bound vs constructions).

Risks: - Some bound-meeting codes may not be unique up to symmetry (multiple non-equivalent codes meeting the same bound). - Classification may already be considered solved by coding theorists; the GC framing adds clarity but not new theorems.

Recommended action: Apply GC retrospectively to coding theory results. Useful as a methodology demonstration, not necessarily new theorems.

Target C: Gauge Anomaly Cancellation

Statement: Which gauge groups + matter representations satisfy all anomaly cancellation conditions in 4D?

GC-amenability: HIGH (rating 1-2). - Finite candidate space: irreducible representations of simple Lie groups are classifiable (Cartan). - Independent bounds: anomaly polynomial vanishing (algebraic constraint) vs gauge group structure (representation theory). - Uniqueness: anomaly-free combinations are highly constrained.

Status: For Standard Model gauge group $SU(3) \times SU(2) \times U(1)$, anomaly cancellation is famously realized by one full generation of fermions. GUT extensions ($SU(5)$, $SO(10)$, E_6) have their own anomaly constraints, mostly satisfied automatically.

Key constraint pair: anomaly polynomial = 0 (upper bound on allowed structure) vs realizability by simple Lie group reps (lower bound).

BST connection: This is exactly the kind of constraint that BST's framework should illuminate. The Standard Model gauge group is forced by BST integers ($m_s = N_c = 3$ forces $SU(3)$; B_2 root system forces additional structure). If anomaly cancellation is forced by the same BST integers, this is a structural unification.

Tractable extensions: - Show: the Standard Model gauge group is the unique anomaly-free gauge group derivable from BST integers. - Extension: which BSM theories are anomaly-allowed? - Holographic anomaly matching (AdS/CFT): BST analog.

Risks: - Anomaly cancellation is well-studied. BST may not produce genuinely new results, only new framings. - The space of allowed gauge groups is small but the space of allowed matter representations is large; GC may be too coarse to navigate.

Recommended action: Investigate connection between BST integer constraints and anomaly polynomial vanishing. Could be a short BST paper.

Target D: Specific Langlands Functoriality Cases

Statement: For specific automorphic representations, prove the predicted functorial transfer (lift to larger group).

GC-amenability: MODERATE-HIGH (rating 2). - Finite candidate space: For fixed source and target groups, automorphic transfers are heavily constrained. - Independent bounds: trace formula on source (lower) vs trace formula on target (upper). Functorial transfer matches them. - Uniqueness: transfers are unique when they exist.

Status: Many specific cases proved (Wiles modularity is one). The general Langlands program is not GC-amenable, but individual transfers often are.

Key constraint pair: trace formula identities on different groups, related by the conjectural functoriality.

Tractable extensions: - Functorial transfers between specific small groups ($GL(2) \rightarrow GL(3)$ is done by Kim-Shahidi; $GL(2) \rightarrow GL(4)$ is open in general). - Specific endoscopic transfers (Arthur's program). - Theta correspondences (BST's main tool — already implicit in BST's BSD work).

Risks: - The Langlands community has its own tools (trace formula, p-adic methods) that may be more efficient than GC. - Individual cases may be solved via methods that don't generalize to a GC framework.

Recommended action: BST's theta correspondence work (BSD, RH) IS already a GC-style attack on specific Langlands cases. Extending to other groups is incremental research, not a new program.

Tier 2: Tractable but Substantial (Rating 2-3)

Target E: Calabi-Yau Moduli at Fixed Hodge Numbers

Statement: For fixed $(h^{\{1,1\}}, h^{\{2,1\}})$, classify Calabi-Yau threefolds up to deformation.

GC-amenability: MODERATE (rating 2-3). - Finite candidate space at fixed Hodge numbers (often a few moduli families). - Independent bounds: Hodge numbers (combinatorial constraint) vs explicit constructions (toric, complete intersection, etc.). - Uniqueness: within a Hodge-data family, deformation classes are finite.

Status: Partial classifications exist (toric CY, weighted projective hypersurfaces). The full classification is open.

Key constraint pair: Hodge numbers + Euler characteristic + intersection numbers.

BST connection: CY 3-folds appear in string compactifications. BST may provide constraints from the 10D bulk (D_{IV}^5) onto admissible CY structures. If physical compactifications correspond to BST-allowed CYs, this is a constraint program.

Tractable extensions: - Specific Hodge profiles (e.g., $(h^{\{1,1\}}, h^{\{2,1\}}) = (1, 101)$ for the quintic 3-fold). - Constructive CY families with explicit moduli. - CY 3-folds appearing in F-theory compactifications.

Risks: - The classification has been open for decades; algebraic geometers have made limited progress. - BST's contribution may be in selecting which CYs are physical, not in classifying all CYs.

Recommended action: Frame BST's potential CY contribution as "physical CYs are BST-constrained subset of all CYs." Pursue after methodology paper lands.

Target F: Optimization Landscapes with Geometric Structure

Statement: For optimization problems with symmetric structure, identify the unique optimum via constraint matching.

GC-amenability: MODERATE (rating 2). - Finite candidate space when symmetry is high (extrema lie at fixed points of symmetry group). - Independent bounds: duality (LP, SDP) provides upper bounds; constructions provide lower bounds. - Uniqueness: convex problems have unique optima; structured non-convex problems often have countably many.

Status: Convex optimization is fully understood. Non-convex with structure (e.g., semidefinite programming relaxations, sum-of-squares hierarchies) is active research.

Key constraint pair: dual bound vs primal construction.

Tractable extensions: - Polynomial optimization via SOS (Lasserre hierarchy). - Quantum optimization (variational quantum eigensolvers). - Algorithmic game theory equilibria.

Risks: - Optimization is a vast field. GC may rediscover known results. - The "constraint" framing may add little to existing duality theory.

Recommended action: Limited. Apply GC to specific high-symmetry optimization problems where uniqueness is novel.

Target G: Quantum Error Correction Code Bounds

Statement: For quantum codes with parameters $[[n, k, d]]$, find tightest bounds and codes that meet them.

GC-amenability: HIGH (rating 2). - Finite candidate space: quantum codes of fixed length, dimension, and distance over Pauli operators. - Independent bounds: quantum Singleton (Knill-Laflamme), quantum Hamming, quantum LP bounds vs CSS construction, stabilizer constructions. - Uniqueness: codes meeting tight bounds.

Status: Active research. Steane code, Bacon-Shor, Kitaev codes are well-studied. Quantum LDPC codes and topological codes have rapidly evolving bounds.

Key constraint pair: quantum-specific bounds (from no-cloning, complementarity) vs stabilizer constructions.

Tractable extensions: - Topological codes (surface code, color code) — finite parameter classification. - Quantum LDPC bounds (recent breakthrough by Panteleev-Kalachev). - Magic state distillation thresholds.

Recommended action: Active field. GC could provide methodology framing for new results. Worth a short paper.

Tier 3: Hard, Multi-Year Programs (Rating 3-4)

Target H: 4-Manifold Smooth Structures (Restricted)

Statement: Classify smooth structures on specific 4-manifold classes (Kähler, Einstein, bounded geometry).

GC-amenability: MODERATE in restricted form (rating 3-4).

Already analyzed in GC-3. Restricted versions are tractable; full classification is not. See GC-3 for details.

Recommended action: Address as scoping in methodology paper. Not a near-term program.

Target I: Mirror Symmetry as Uniqueness

Statement: Show that mirror pairs of Calabi-Yau manifolds are uniquely determined by their Hodge data swap.

GC-amenability: MODERATE (rating 3). - Finite candidate space for each Hodge profile. - Independent bounds: A-model (symplectic geometry) vs B-model (complex geometry) — mirror identifies them. - Uniqueness: for many CY pairs, mirror is unique.

Status: Mirror symmetry is conjectural in many cases (proven for specific families: quintic-mirror by Givental, Lian-Liu-Yau).

Key constraint pair: A-model invariants vs B-model invariants under mirror map.

Recommended action: Long-term research program. BST may contribute via D_{IV}^5 's Hodge structure $\rightarrow$ preferred mirror partners. Not for near-term.

Target J: Quantum Gravity Arena Selection

Statement: Which manifold supports a consistent quantum theory of gravity?

GC-amenability: SPECULATIVE (rating 4).

- Candidate space: bounded symmetric domains, Calabi-Yau, AdS spaces, others.
- Independent bounds: Wightman/Haag-Kastler axioms (upper) vs constructive QG (lower) — no construction exists.
- Uniqueness: BST claims D_{IV}^5 ; AdS/CFT claims $AdS_5 \times S^5$; loop QG claims spin networks. Multiple frameworks compete.

Status: Quantum gravity is a 70+ year open problem. No constructive QG exists in 4D.

Key constraint pair: in BST's framing, the same five integers that force D_{IV}^5 should also force QG to live there. The constraint pair would be: spectral structure + symmetry + finite mass gap.

BST connection: BST already places physics on D_{IV}^5 . Extending to QG would be the most ambitious BST claim — that gravity emerges from the curvature of D_{IV}^5 itself.

Risks: - Constructive QG has resisted 70 years of attack. - BST's contribution to QG is currently speculative (boundary conditions, eigentones). - The competition (string theory, loop QG, asymptotic safety) is well-entrenched.

Recommended action: Long-term BST research program (BST-GR, separate from BST-SR). Not for near-term. The methodology paper should mention this as the largest open application.

Tier 4: Research Frontier (Rating 5)

Target K: Full Langlands Program

Statement: For every reductive group G and every automorphic representation, prove the predicted Galois representation correspondence.

GC-amenability: LOW (rating 5).

The general Langlands program is not GC-amenable. It is a structural correspondence (Wiles-style), not a constraint problem. Individual cases (Target D above) are GC-tractable; the general program is not.

Recommended action: Do not target. Acknowledge as out-of-scope.

Target L: Hodge Conjecture in Full Generality

Statement: For every smooth projective variety, every rational Hodge class is algebraic.

GC-amenability: PARTIAL (rating 4-5).

Already analyzed in BST's Hodge program. BST proves Hodge for D_{IV}^5 Shimura varieties (Paper H1) and extends via Kuga-Satake (Paper H2). The full generality is beyond BST's current reach (see Hodge papers' honest scope).

Recommended action: BST already addresses what's tractable. The remaining cases are not GC-amenable in current formulation.

Priority Recommendations

For the methodology paper (GC-9) and post-submission research programs:

Highest priority (start now or after Clay submissions):

1. Target A (Sphere Packing extensions) — clean GC application, builds on Viazovska, low risk
2. Target C (Anomaly Cancellation) — direct BST connection, short paper potential
3. Target B (Error-Correcting Code Boundaries) — retrospective application, methodology paper case study

Medium priority (after methodology paper):

4. Target D (Specific Langlands Cases) — extends BST's existing BSD/RH work
5. Target G (Quantum Error Correction) — active field, new results possible
6. Target E (Calabi-Yau at Fixed Hodge Numbers) — long-term, connects to string compactifications

Long-term research programs (BST-GR territory):

7. Target J (Quantum Gravity) — BST's most ambitious claim, long-term commitment
8. Target I (Mirror Symmetry) — multi-year program

Do not target (current formulation):

9. Target K (Full Langlands) — wrong shape for GC
10. Target L (Full Hodge) — BST already does what's tractable

Implications for the Methodology Paper

GC-9 should include an applications section organized as:

“Where GC Works” (case studies of solved problems): - Poincaré (Perelman) — already analyzed in GC-2 - Viazovska sphere packing (dim 8, 24) — established - BST 7 Clay problems — this work - Four-Color (BST version) — already analyzed in GC-4

“Where GC Plausibly Works” (open targets, near-term): - Sphere packing in other dimensions - Anomaly cancellation - Specific Langlands cases - Error-correcting code boundaries

“Where GC Might Work” (longer-term): - Calabi-Yau moduli at fixed Hodge data - Quantum gravity arena selection (speculative) - Mirror symmetry uniqueness

“Where GC Does Not Work” (honest scope): - General Langlands program - Smooth 4-manifold classification (in full generality) - Existence-only theorems (Goldbach, bounded gaps, Green-Tao type) - Probabilistic/density results

This three-tier structure (works / plausibly works / doesn’t work) keeps the methodology paper’s claims calibrated. Don’t oversell; map the boundary honestly.

Common Pattern Across Targets

Reviewing all twelve targets, the GC method works when:

1. Constraints come from outside the structure being classified (curvature from ambient manifold, anomaly polynomial from gauge structure, etc.)
2. The constraints are independent (anomaly + representation theory, LP bound + lattice construction, etc.)
3. The candidate space is enumerable (representations of Lie groups, lattices in fixed dim, etc.)
4. Bounds meet exactly at the answer (Viazovska’s magic functions, BST integer squeeze, anomaly-free reps, etc.)

GC fails when:

1. No external constraint exists (problem is purely internal, like full smooth 4-manifold classification)
2. Existence is the question, not uniqueness (Goldbach, twin primes)
3. The candidate space is uncountable (general moduli, smooth 4-manifolds in isolation)
4. Results are statistical (Green-Tao, prime gaps)

The GC method is a specific tool for a specific shape of problem. Recognizing the shape is half the work; applying the constraint pair is the other half.

References

- GC-4: notes/BST_GC4_Survey_Solved_Problems.md — Grace’s survey of solved problems
- GC-5: notes/BST_GC5_Five_Step_Methodology.md — methodology formalization
- GC-6: notes/BST_GC6_Dimension_Ladder.md — dimensional classification

- Viazovska, M. (2017): “The sphere packing problem in dimension 8.” *Annals of Math.* 185.
 - Cohn, H. (2002): “New upper bounds on sphere packings.” *Annals of Math.* 157.
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Revision History

- v0.1 (May 12, 2026): Initial assessment. Twelve targets rated for tractability. Three-tier organization for methodology paper applications section. Honest scope: GC works for problems with specific shape (uniqueness from external independent constraints on enumerable candidates), fails for problems without that shape. Grace’s portion catalogs the formal statements; this portion assesses tractability and prioritization.